

SOLUTION ARCHITECTURE OVERVIEW

The solution architecture establishes a multi-tier, modular design within the ServiceNow platform. It decouples the student-facing presentation layer from the background classification engine, database schema, security controls, and change tracking mechanisms.

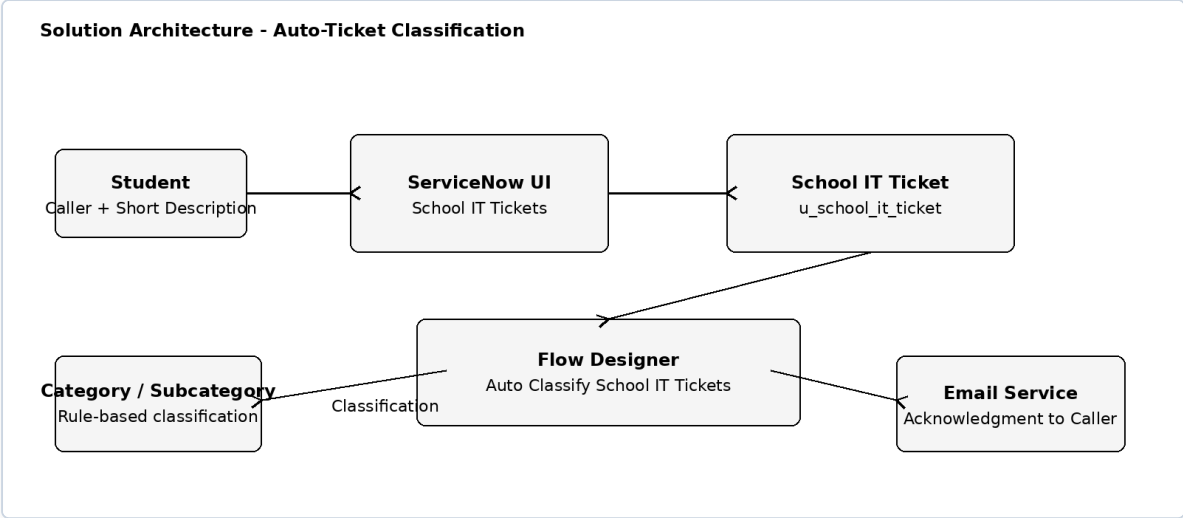


Figure 1: Solution architecture and data flow of the School IT ticket classification system.

ARCHITECTURE COMPONENTS BREAKDOWN

| Component Layer      | Role in Solution Architecture                                                              | ServiceNow Technology            |
|----------------------|--------------------------------------------------------------------------------------------|----------------------------------|
| Presentation Layer   | Provides intuitive ticket creation forms and list views for requesters and technicians.    | Next Experience UI / Workspace   |
| Data Schema Layer    | Stores structured records: Number, Caller, Category, Subcategory, Descriptions, State.     | Custom table: u_school_it_ticket |
| Automation Logic     | Evaluates short description keywords upon record creation and auto-populates categories.   | ServiceNow Flow Designer         |
| Notification Service | Generates and delivers instant acknowledgment emails containing ticket reference details.  | ServiceNow Email Engine / SMTP   |
| Security & Access    | Enforces role-based permissions governing who can create, view, and update ticket records. | ServiceNow ACLs and User Roles   |
| Change Management    | Packages all schema, form layouts, flows, and choice configurations for release control.   | ServiceNow Update Sets (XML)     |

IMPLEMENTATION EVIDENCE - FORM AND LOGIC INTEGRATION

The connection between the data collection interface and the Flow Designer execution engine is shown below.

The screenshot shows the 'School IT Ticket' form in ServiceNow. The form includes the following fields:

- Number:** TK10001012
- Caller:** System Administrator
- Short description:** Wi-Fi not working in library
- Description:** (Empty text area)
- Category:** Network
- Subcategory:** Wi-Fi
- State:** -- None --
- Assigned Group:** (Empty dropdown)
- Assigned to:** (Empty dropdown)

Buttons for 'Update' and 'Delete' are located at the bottom left of the form.

Figure 2: School IT Ticket form showing user-facing ticket submission fields.

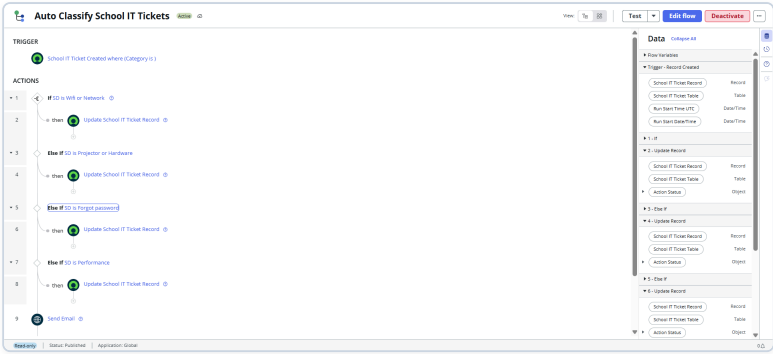


Figure 3: Published Auto Classify School IT Tickets Flow Designer workflow showing condition logic.

IMPLEMENTATION EVIDENCE - EXECUTION AND DEPLOYMENT PACKAGING

The live classification outcome and the completed deployment update set are shown below.

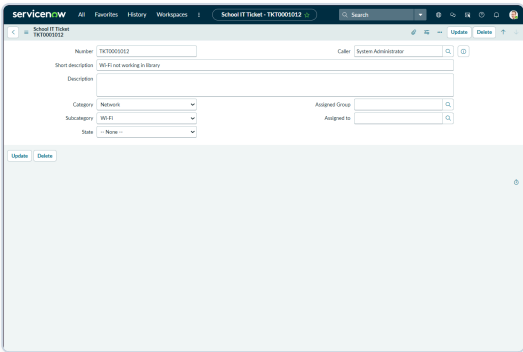


Figure 4: School IT Ticket showing the result of automatic classification (Category: Network, Subcategory: Wi-Fi).

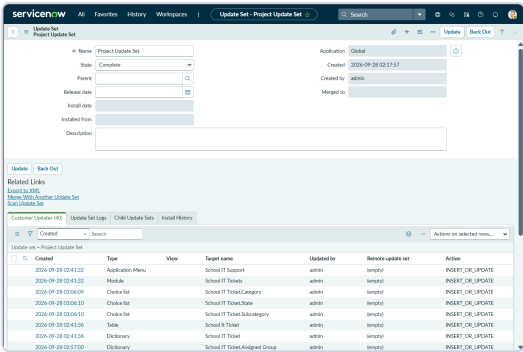


Figure 5: Completed Project Update Set containing captured customizations and XML export option.